

GBM Confidence-Enhanced Survival Inference Project

Detailed Student Protocol with Clinical Questions and Techniques

Step 2: Kaplan-Meier Survival Analysis

Clinical Question: Which individual mutations are associated with clinically meaningful differences in overall survival?

Technique: Kaplan-Meier Survival Analysis + Log-Rank Test

What is this technique?

Kaplan-Meier estimates the probability of survival over time. The log-rank test compares survival curves between mutated and non-mutated groups.

Student Instructions:

1. For each gene (ATRX, TERT, BRAF, EGFR, TP53, IDH1, IDH2), split patients into mutated and non-mutated groups.
2. Generate Kaplan-Meier curves.
3. Compute median survival.
4. Perform a log-rank test.
5. Save one figure per mutation.

What to look for:

- Clear separation between curves.
- Significant log-rank p-value.
- Meaningful median survival difference.

Expected Output:

Kaplan-Meier curves and median survival table.

Step 3: Cox Proportional Hazards Modeling

Clinical Question: Which mutations provide independent prognostic information?

Technique: Cox Proportional Hazards Model

What is this technique?

Cox regression estimates the hazard ratio (HR), which quantifies how much a mutation increases or decreases the risk of death.

Student Instructions:

1. Fit an unadjusted Cox model for each mutation.
2. Fit an adjusted Cox model using age, sex, and available covariates.
3. Record HR, 95% CI, and p-values.
4. Create a forest plot summarizing all mutations.

What to look for:

- $HR > 1$ indicates increased risk.
- $HR < 1$ indicates protective effect.
- Compare adjusted vs. unadjusted HRs.

Expected Output:

Hazard ratio table and forest plot.

Step 4: Propensity Score Enhanced Analysis

Clinical Question: Are mutation-survival associations robust after balancing patient characteristics?

Technique: Propensity Score Modeling + IPTW + Weighted Cox Regression

What is a propensity score?

The probability that a patient belongs to the mutation group based on age, sex, stage, MGMT, and other available covariates.

Student Instructions:

1. Use logistic regression to estimate propensity scores.
2. Plot propensity score overlap between mutation groups.
3. Compute IPTW weights.
4. Evaluate balance using Standardized Mean Differences (SMD).
5. Fit an IPTW-weighted Cox model.

What to look for:

- Good overlap between groups.
- $SMD < 0.10$ after weighting.
- Similar HRs between adjusted Cox and IPTW Cox.

Expected Output:

Propensity overlap plot, balance table, weighted hazard ratios.

Step 5: Co-Mutation Analysis

Clinical Question: Do combinations of mutations define distinct survival phenotypes?

Technique: Co-Mutation Survival Analysis

What is this technique?

Instead of studying a single mutation, evaluate the combined effect of two mutations.

Student Instructions:

1. Analyze TP53+EGFR, TP53+IDH1, ATRX+IDH1, and TERT+EGFR.
2. Create four groups:
 - Neither mutation
 - Mutation A only
 - Mutation B only
 - Both mutations
3. Generate Kaplan-Meier curves.
4. Fit adjusted Cox models.

What to look for:

- Whether the co-mutated group has substantially different survival.
- Whether co-mutations provide stronger prognostic information than single mutations.

Expected Output:

Four-group Kaplan-Meier curves and co-mutation HR table.

Step 6: Survival Machine Learning

Clinical Question: Which mutations contribute most strongly to survival prediction?

Technique: Random Survival Forest (RSF) and Optional XGBoost Survival

What is this technique?

Machine-learning survival models learn complex relationships between mutations and survival outcomes.

Student Instructions:

1. Build a Random Survival Forest model.
2. Evaluate using Concordance Index (C-index).
3. Compute variable importance.
4. Optionally build XGBoost Survival.
5. Rank mutations by importance.

What to look for:

- Higher C-index indicates better prediction.
- Consistently important mutations.

Expected Output:

Performance table and mutation ranking plot.

Step 7: Confidence Framework

Clinical Question: Which biomarkers are sufficiently reproducible to be trusted clinically?

Technique: Evidence Integration Framework

Student Instructions:

1. Collect results from KM, Cox, IPTW, and ML analyses.
2. Compare direction and significance across methods.
3. Assign a confidence score.
4. Classify biomarkers as High, Moderate, or Low confidence.

What to look for:

- Consistency across methods.
- Stable hazard ratios.
- Biological plausibility.

Expected Output:

Confidence ranking table.